

llmXive Automated Review of WBench: A Comprehensive Multi-turn Benchmark for Interactive Video World Model Evaluation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a comprehensive benchmark with extensive quantitative claims. However, several specific factual assertions require verification to ensure they do not overstate the evidence or contain minor statistical inaccuracies.

First, the claim in Section 5.2 that navigation performance is “largely independent” with a “correlation 0” needs precise verification. The critical elements list includes values like 0.05 and 0.12 which may correspond to these correlations. If the actual correlation is 0.05 or 0.12, describing it as “0” or “independent” is a slight overstatement. The text should be adjusted to reflect the exact coefficient (e.g., “weakly correlated”) or explicitly state that the correlation is statistically insignificant if a p-value is available.

Second, the assertion in Section 5.2 that “four =1.00” in the human-auto alignment analysis is statistically suspect. Achieving a perfect Spearman rank correlation of 1.00 across human pairwise judgments (which inherently contain noise) is extremely rare. It is highly likely these values are rounded (e.g., 0.996 or 0.998). The text should be corrected to reflect the actual precision (e.g., “0.99”) to avoid implying an impossible level of agreement.

Third, the central claim that “no single model excels on all dimensions” is generally supported by the data, but the phrasing in the abstract and conclusion could be sharper. Table 4 shows specific models leading in specific dimensions (e.g., LingBot-World in Consistency/Physical, Seedance in Setting Adherence). The text should explicitly clarify that no model leads in *all five* dimensions simultaneously, rather than the current phrasing which could be misread as “no model leads in any dimension.”

Finally, the statistical breakdown in Section 4.1 (62% FPP, 38% TPP) should be checked for rounding consistency. If these are rounded figures, the text should use “approximately” to maintain factual rigor. The subject distribution (64% humans, 9% animals, etc.) also lists an “others” category (10%) which should be explicitly accounted for in the total to ensure the percentages sum correctly to 100%.

Action items.

- **[writing]** Section 5.2 claims navigation correlation is ‘0’. Verify exact value (e.g., 0.05) in Fig 3a. If non-zero, change to ‘weakly correlated’ to avoid overstatement.
- **[science]** Section 5.2 states ‘four =1.00’ for human alignment. Perfect correlation is statistically improbable with 400 annotators. Verify if this is rounding (e.g., 0.996) and adjust text to ‘0.99’.
- **[writing]** Abstract claims ‘no single model excels on all dimensions’. Clarify this means no model leads in all five simultaneously, as Table 4 shows leaders in individual dimensions.
- **[writing]** Section 4.1 states ‘62% FPP, 38% TPP’. Verify if these are exact or rounded. If

rounded, add ‘approximately’ to ensure factual precision.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear visual overview of the benchmark’s components and taxonomy, but the caption contains significant text omissions (missing benchmark name) and the evaluation radar chart lacks a numerical scale.

Figure 2

The figure effectively visualizes correlations and deviations, but the Z-score heatmap in panel (c) uses a color scale that clips extreme values, and the y-axis labels are illegible due to overlap.

Figure 3

The figure effectively visualizes performance trends across turns, but the ‘Perspective Switching’ panel contains a confusing mismatch between the legend symbol and the plotted baseline data, and the caption lacks specific metric definitions.

Figure 4

The figure effectively visualizes the correlation between human and automated scores with clear axes and a legend. However, the caption contains significant formatting errors where mathematical symbols and values are rendered as raw LaTeX code, obscuring the reported statistics.

Figure 5

Figure 5 is a visually clear thumbnail gallery, but its caption is grammatically incomplete and missing the benchmark name after ‘in’, reducing clarity.

Figure 6

The figure provides a clear visual gallery of scene types, but the caption is factually inconsistent with the layout, incorrectly describing the number of categories per row and the nature of the

image pairs.

Figure 7

Figure 7 effectively displays representative initial frames for the seven rendering styles listed in the caption (realistic, anime, cartoon, oil painting, ink wash, flat, and pencil sketch). The visual examples are clear, distinct, and directly support the claim of a style gallery.

Figure 8

Figure 8 effectively illustrates the three perspective categories (disembodied first-person, embodied first-person, and third-person) with clear visual examples. The caption accurately describes the grouping and content, and the image is legible with no missing labels or controls.

Figure 9

Figure 9 is a clear and well-organized gallery of images categorized by subject type (Human, Animal, Vehicle, Robot, Other). The visual layout effectively demonstrates the diversity of scenes and styles covered by the benchmark, and the caption accurately describes the content shown.

Figure 10

The figure provides a visual example of perspective switching but fails to deliver the comprehensive taxonomy promised in the caption, showing only two cases instead of the three distinct sub-types described. Additionally, the text labels are somewhat small and could be improved for readability.

Figure 11

Figure 11 effectively illustrates the navigation action definitions for both first-person and third-person perspectives. The visual mapping of WASD keys and arrow keys to physical camera movements is clear, and the caption provides necessary context regarding the source of the frames.

Figure 12

The figure presents navigation action and trajectory distributions but fails to visualize the ‘scene type’ and ‘control interface’ distributions promised in the caption. Additionally, the third subplot has an incorrect label (‘(b)’ instead of ‘(c)’), and the first subplot lacks explicit context linking the labels to navigation controls.

Action items.

- **[writing]** Figure 1: The caption contains multiple instances of missing text where the benchmark name should appear (e.g., ‘Overview of .’, ‘cases in .’, ‘covered by :’). The figure itself also uses the abbreviation ‘PS’ without defining it in the caption.

- **[science]** Figure 1: The radar chart in panel (d) lacks a visible scale or grid values, making it impossible to determine the magnitude of the scores for the plotted models.
- **[science]** Figure 2(c): The heatmap contains values exceeding the theoretical range of a Z-score (e.g., +1.9, -1.9), yet the colorbar scale is fixed to [-1.0, 1.0]. This causes the most extreme values to be clipped to the maximum/minimum colors, obscuring the true magnitude of the deviations.
- **[writing]** Figure 2(c): The y-axis labels are rotated 90 degrees and overlap significantly, making them difficult to read; horizontal alignment or increased spacing is needed.
- **[science]** Figure 3: The ‘Perspective Switching’ panel shows a red ‘X’ and a dashed line for the ‘9 text-driven models’ baseline, but the legend entry for this group is a solid black square. This visual mismatch makes the baseline data illegible and confusing.
- **[writing]** Figure 3: The caption ‘Per-turn performance degradation’ is too brief; it should explicitly state that the y-axis represents the evaluation score (e.g., 0-100) to clarify the metric being degraded.
- **[writing]** Figure 4: The caption contains LaTeX formatting artifacts (‘Spearman \$’, ‘0.94’, ‘\$ = 1.00’) instead of the symbol ‘ ρ ’ or the word ‘correlation’, making the metric and values unreadable.
- **[writing]** Figure 4: The legend at the top is not explicitly labeled as ‘Models’ or ‘Methods’, though the context implies it; adding a label would improve clarity.
- **[writing]** Figure 5 caption contains a missing reference: ‘all cases in .’ lacks the benchmark name (likely ‘WBench’).
- **[writing]** Figure 5 caption is incomplete: ‘all cases in .’ should specify the dataset or benchmark name.
- **[writing]** Figure 6: The caption claims ‘Two categories per row are presented,’ but the layout displays four distinct categories per row (Nature, Urban, Indoor, Workspace in the top half; Fantasy, Sports in the bottom half), creating a direct contradiction between the text and the visual structure.
- **[writing]** Figure 6: The caption states each category is shown as a ‘photorealistic/stylized pair,’ yet the ‘Nature’ and ‘Urban’ rows contain multiple disparate images (e.g., penguin, bamboo, city street, neon city) rather than a single paired comparison, making the description inaccurate for the majority of the content.
- **[science]** Figure 10: The caption claims to showcase ‘same-subject switches, multi-subject switches, and scope-mode transitions,’ but the image only displays two examples (scope-to-FPP and FPP-to-FPP) and lacks the promised variety of sub-types.
- **[writing]** Figure 10: The text labels ‘TPP -> FPP’, ‘FPP -> FPP’, etc., are rendered at a resolution that is difficult to read and may be illegible in smaller formats.

- **[writing]** Figure 12: The caption claims the figure shows distributions across ‘direction, scene type, and control interface’, but the figure only displays distributions for ‘direction’ (Atomic Distribution) and ‘trajectory type’ (Trajectory/Mix Distribution). The ‘scene type’ and ‘control interface’ distributions mentioned in the caption are missing from the visual.
- **[writing]** Figure 12: The third subplot is labeled ‘(b) Mix Distribution’, but it should be labeled ‘(c)’ to follow the sequential ordering of (a) and (b) in the preceding subplots.
- **[science]** Figure 12: The ‘Atomic Distribution’ chart (a) labels segments with single letters (W, S, A, D) and directions (up, down, left, right) but does not explicitly state that these represent navigation key presses or actions, which is critical context for a ‘navigation test case’ distribution.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript is dense with domain-specific acronyms and abbreviations that hinder accessibility for readers outside the immediate niche of generative video and robotics. While the technical precision is high, the text frequently assumes prior knowledge of terms that should be explicitly defined or replaced with plain English.

Specifically, the term “6-DoF” appears in the Abstract and Introduction without definition. While standard in robotics, a general computer science reader may not immediately grasp “Degrees of Freedom.” Similarly, “nATE” is introduced in Section 5 and used throughout the Appendix without ever being spelled out as “normalized Average Translation Error.” This forces the reader to guess the metric’s nature or search external literature.

The acronym “DiT” (Diffusion Transformer) is used repeatedly in the Experiments section (e.g., describing LTX, LongCat, Kairos) without a prior definition. While “Transformer” is well-known, the specific “Diffusion” variant is a distinct architectural choice that warrants a brief explanation or full spelling at first use.

Furthermore, the table captions and text rely heavily on “FPP” and “TPP” for First-Person and Third-Person perspectives. These are gaming/VR conventions that should be written out in full in the main text or table headers to ensure clarity for a broader academic audience. Finally, “VLM” is used extensively to describe the evaluation judges (e.g., in Section 5 and Appendix C) but is never defined as “Vision-Language Model.”

To improve readability and inclusivity, the authors should perform a pass to ensure every acronym is defined at its first occurrence and consider replacing highly specific jargon (like “DoF” or “DiT”) with descriptive phrases where the context allows, or at least providing a brief parenthetical explanation.

Action items.

- **[writing]** The manuscript is dense with domain-specific acronyms and abbreviations that hinder accessibility for readers outside the immediate niche of generative video and robotics. While the technical precision is high, the text frequently assumes prior knowledge of terms that should be explicitly defined or replaced with plain English. Specifically, the term “6-DoF” appears in the Abstract and Introduction without definition. While standard in robotics, a general computer science reader may not immediat

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a comprehensive benchmark, but several logical gaps exist in the interpretation of experimental results.

First, in Section 5.2 (“Results Summary”), the authors state that “navigation is largely independent (correlation 0) from quality, consistency, and physics.” However, the same section explicitly reports that “Physical scores correlate with video quality ($r=0.84$).” Logically, if Variable A (Navigation) is uncorrelated with B (Quality) and C (Physics), and B and C are highly correlated, the statement “independent from quality, consistency, and physics” is technically correct but potentially misleading without clarifying that Navigation is independent of the *joint* distribution of Quality and Physics. The text implies a total decoupling that might be better phrased as “Navigation performance does not predict the Quality-Physics cluster.”

Second, the causal claim in the Conclusion (“physical correctness follows rendering quality”) is not fully supported by the stated evidence. The paper establishes a strong correlation ($r=0.84$) between Physical scores and Video Quality. Correlation does not imply causation; it is equally plausible that models with better physical understanding generate higher-quality videos, or that a third factor (e.g., model scale) drives both. The paper does not provide a mechanism (e.g., “better rendering allows the VLM judge to detect physics violations more accurately”) to support the directional claim that quality *causes* physical correctness.

Third, the analysis of “Per-turn degradation” (Section 5.2) notes that perspective switching scores “remain flat” while navigation degrades. However, the dataset statistics (Section 4.1) indicate perspective switching comprises only 6% of interactions. The stability of the score could be a result of insufficient statistical power (high variance in a small sample) rather than a genuine robustness of the capability. The paper draws a strong comparative conclusion (“remains flat” vs. “drops 33 points”) without addressing the sample size disparity, which weakens the logical validity of the comparison.

Finally, the definition of the “Physical” dimension relies heavily on VLM scoring (Causal Fidelity). The paper claims these scores correlate with human judgments (Fig 3), but the logical link between “VLM scoring” and “Physical correctness” assumes the VLM possesses ground-truth physics knowledge. The paper does not logically bridge the gap between “VLM agreement

with humans” and “VLM correctness regarding physics,” leaving a potential circularity in the evaluation of the Physical dimension itself.

Action items.

- **[writing]** The paper presents a comprehensive benchmark, but several logical gaps exist in the interpretation of experimental results. First, in Section 5.2 (“Results Summary”), the authors state that “navigation is largely independent (correlation 0) from quality, consistency, and physics.” However, the same section explicitly reports that “Physical scores correlate with video quality ($r=0.84$).” Logically, if Variable A (Navigation) is uncorrelated with B (Quality) and C (Physics), and B and C are highly

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the state of interactive world models that slightly exceed the immediate evidence provided in the experimental results.

First, the central conclusion that “no single model excels on all dimensions” is a valid observation of the current landscape, but the phrasing suggests a definitive, universal law rather than a snapshot of the specific 20 models tested. While Table 1 (e002) supports the lack of a single “winner,” the paper does not rigorously test the statistical significance of this “no dominance” claim across the entire space of possible models. It is a reasonable summary, but the language could be tempered to “no single model *among those evaluated* excels...” to avoid over-generalization.

Second, the claim that “Physical scores correlate with video quality” (Section 5.2, $r=0.84$) is presented as a general finding. However, the correlation analysis in Figure 5a is based on the navigation split (158 cases) and potentially a subset of models. The paper does not explicitly state whether this high correlation holds when comparing the full set of 289 cases or across all 20 models. If the correlation is driven primarily by the text-driven models (which tend to have high quality and high physics scores due to their generation nature) and not the action-conditioned models, then generalizing this as a fundamental property of world models is an overreach. The authors should clarify the exact subset of data and models used to derive this correlation coefficient.

Third, the assertion that “navigation is decoupled from other abilities” relies on a correlation of approximately zero. However, the evaluation protocol differs: navigation is evaluated on a specific subset (158 cases) for all models, while other metrics (like Setting Adherence) are evaluated on the full set (289 cases) for text-driven models. Comparing correlations across these different evaluation scopes without explicitly addressing the potential confounding factor of dataset size or composition could lead to an over-interpreted conclusion about the independence of these capabilities. The authors should clarify if the correlation analysis was performed on a

strictly overlapping subset of data.

Finally, the statement that “Open-source models achieve competitive scores” (Conclusion) is slightly misleading given that the top performers in key categories (Wan 2.7, Kling 3.0) are API-only. While open models like LongCat and LingBot perform well, the phrasing implies a broader trend of open-source parity that the data does not fully support. Qualifying this to “some open-source models achieve competitive scores” would be more accurate.

These issues are primarily matters of precision in language and scope rather than fatal flaws, but addressing them will prevent the paper from over-claiming the universality of its findings.

Action items.

- **[writing]** The claim ‘no single model excels on all dimensions’ (Abstract) is an over-generalization. Table 1 shows leaders vary, but the paper lacks statistical proof that no model is second-best in all others. Clarify if this is a qualitative summary or a robust statistical finding.
- **[science]** The assertion ‘Physical scores correlate with video quality’ ($r=0.84$) does not specify the model subset. If driven only by text-driven models, generalizing this to all world models is an overreach. Specify the exact subset used for this correlation.
- **[science]** The claim ‘navigation is decoupled’ relies on correlations from different evaluation subsets (158 vs 289 cases). Comparing across non-overlapping data risks over-interpreting independence. Clarify the data overlap used for the correlation analysis.
- **[writing]** The statement ‘Open-source models achieve competitive scores’ is misleading as top performers (Wan 2.7, Kling 3.0) are API-only. Qualify this to ‘some open-source models’ to avoid implying general parity with closed models.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a comprehensive benchmark for interactive video world models. From a safety and ethics perspective, the paper is generally sound in its intent to evaluate model capabilities rather than generate harmful media. However, three specific areas require clarification to ensure full compliance with ethical research standards.

First, regarding **Dual-Use and Content Safety**: The dataset includes interaction types such as “combat” and “subject action” (Section 5, Appendix e002). While the paper states in the Broader Impact section (Appendix e002) that it mitigates risks by releasing only evaluation tools, it does not explicitly address whether the 289 test cases or the prompt templates include scenarios that could be interpreted as generating violence, self-harm, or other harmful content. Given the rapid advancement of generative video, there is a risk that the benchmark itself could be repurposed to stress-test models for generating harmful imagery. The authors should

explicitly state whether a content safety filter was applied during the curation of the 289 cases and whether any prompts were excluded due to safety concerns.

Second, regarding **Human Subject Research**: The evaluation of the benchmark’s alignment with human preference involved 400 annotators performing 13,515 pairwise comparisons (Appendix e002, Section “Human-Preference Annotation Platform”). The manuscript does not mention Institutional Review Board (IRB) approval or ethical oversight for this crowdsourcing activity. Standard ethical guidelines require that such studies be reviewed to ensure informed consent, fair compensation, and protection of annotator well-being (e.g., exposure to potentially disturbing content). The authors must confirm if IRB approval was obtained and provide details on the consent process and compensation rates.

Third, regarding **Data Privacy and Training Data**: The “Visual Plausibility” metric relies on a model fine-tuned on 6,000 videos with human annotations (Appendix e002, Section “Visual Plausibility”). The provenance of these 6,000 videos is not detailed. If these videos contain real-world footage of individuals, the authors must confirm that appropriate consent was obtained for their use in training a public model and that no personally identifiable information (PII) was retained. If the data is synthetic, this should be explicitly stated to avoid ambiguity.

Addressing these points will strengthen the paper’s ethical standing and ensure transparency regarding the safety of the dataset and the welfare of the human participants involved in the evaluation.

Action items.

- **[writing]** The manuscript presents a comprehensive benchmark for interactive video world models. From a safety and ethics perspective, the paper is generally sound in its intent to evaluate model capabilities rather than generate harmful media. However, three specific areas require clarification to ensure full compliance with ethical research standards. First, regarding Dual-Use and Content Safety: The dataset includes interaction types such as “combat” and “subject action” (Section 5, Appendix e002). Whil

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a substantial benchmark with a large dataset (289 cases, 1,058 turns) and extensive evaluation across 20 models. The sample size is sufficient for a comparative analysis of model capabilities. However, the scientific evidence supporting the validity of the proposed metrics requires further clarification.

First, the central claim of high alignment between automated metrics and human judgment (Spearman ρ 0.94, Fig. 5) is critical for the benchmark’s utility. While the number of tasks (13.5K) is large, the review lacks a report on the inter-annotator agreement (IAA) among the 400 human workers. Without IAA, it is impossible to determine if the high correlation is driven

by a consistent human signal or if the automated metrics are simply overfitting to a noisy human consensus. Confidence intervals for the correlation coefficients are also missing.

Second, the “Navigation” metric (NavScore) relies on comparing predicted camera poses (via MegaSaM) against synthetic ground truth. The validity of this metric hinges on the accuracy of the pose estimator on the specific synthetic video styles generated by the models. The paper does not provide a validation of MegaSaM’s performance on this specific domain or a sensitivity analysis of the NavScore to pose estimation errors. If the pose estimator fails on certain generated textures or lighting conditions, the resulting “navigation failure” might be an artifact of the metric rather than the model.

Third, the “Physical” dimension (Causal Fidelity) is evaluated using VLMs (Doubao-Seed-2.0-lite) with specific prompts. While the prompts are provided in the appendix, there is no evidence of the metric’s robustness to prompt variations or its stability across different VLM architectures. The lack of a reported inter-rater reliability between multiple VLM instances or between VLMs and human experts for the physics scores leaves the objectivity of this dimension unverified.

Finally, the degradation analysis (Fig. 4) shows a sharp drop in navigation scores by turn 4. The paper attributes this to model limitations but does not rule out the possibility of error accumulation in the iterative generation protocol or the evaluation pipeline itself. A control experiment isolating the evaluation noise from the generation noise would strengthen this claim.

Action items.

- **[science]** The claim of strong human-auto alignment (Spearman $\rho = 0.94$, Fig. 5) relies on 400 annotators but lacks statistical power analysis or confidence intervals. With 13.5K tasks, the sample size is large, but the variance in human judgments (inter-annotator agreement) is not reported, making the robustness of the correlation claim uncertain.
- **[science]** Navigation accuracy (NavScore) depends on MegaSaM for pose estimation against synthetic GT. The paper does not report the error rate of MegaSaM on the specific synthetic data distribution or validate the GT construction logic. If the pose estimator fails on the generated video styles, the ‘ground truth’ comparison is invalid.
- **[science]** The ‘Physical’ dimension relies heavily on VLMs (Doubao-Seed-2.0-lite) for Causal Fidelity. The prompt engineering and rubric (Appendix) are provided, but no inter-rater reliability (Cohen’s kappa) between VLMs or between VLMs and human experts is reported to validate the metric’s stability against prompt variations.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in WBench is extensive but lacks necessary rigor in uncertainty quan-

tification and error control. While the paper reports 22 sub-metrics across 289 test cases, it presents only point estimates (means) in Tables 1 and 2 (e.g., “Wan 2.7: 91.5”, “Kling 3.0: 91.0”). Without confidence intervals (CIs) or standard errors, it is impossible to determine if these differences are statistically significant or within the noise of the evaluation protocol. Given the multi-turn nature of the data (1,058 turns), the effective sample size for independence assumptions also requires clarification.

Furthermore, the paper performs numerous pairwise comparisons across models and dimensions without mentioning any correction for multiple testing (e.g., Bonferroni, Holm-Bonferroni, or False Discovery Rate). With 22 metrics and 20 models, the probability of false positives is substantial. The claim that “no single model excels on all dimensions” relies on these uncorrected comparisons.

Regarding the human alignment study (Section 5.3, Figure 4), the authors report high Spearman correlations ($\rho \geq 0.94$). However, the statistical power of this claim depends on the number of pairwise comparisons and the distribution of human scores. The text mentions “400 annotators” and “13.5K tasks,” but the specific N used for the correlation calculation is not explicitly stated in the text or figure caption. Additionally, the validation relies on a single aggregate correlation coefficient; a more robust analysis would include Bland-Altman plots or per-dimension error analysis to ensure the automated metrics do not systematically bias high or low scores.

Finally, the Z-score analysis for difficulty factors (Figure 3c) is presented without the underlying variance or standard deviation of the scores used to compute these Z-scores. To ensure reproducibility and statistical validity, the authors should provide the standard deviation of the scores for each dimension and model, and explicitly state the statistical tests used to derive the Z-scores.

Action items.

- **[science]** Report confidence intervals or standard errors for all mean scores in Tables 1 and 2. With 289 cases, point estimates alone are insufficient to assess statistical significance of model differences (e.g., Wan 2.7 vs. Kling 3.0).
- **[science]** Clarify the multiple-comparisons correction method used for the 22 sub-metrics. Without adjustment (e.g., Bonferroni or FDR), the risk of Type I errors in identifying ‘leading’ models is high.
- **[writing]** Specify the sample size (N) and degrees of freedom for the Pearson/Spearman correlations reported in Figure 3. Correlation strength depends heavily on N, which is currently ambiguous for the ‘20 models’ subset.
- **[science]** Provide the distribution (e.g., boxplots or histograms) of the human-annotation scores used to validate the automated metrics. A single Spearman rho value does not reveal if the alignment holds across the full score range or only at extremes.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high standard of academic writing with clear structure and logical flow. The abstract effectively summarizes the benchmark’s scope, and the introduction successfully motivates the need for a unified evaluation standard. The prose is generally concise, and the transition between the dataset description and the evaluation suite is smooth.

However, a few specific areas require minor polishing to ensure maximum clarity for a broad audience. In Section 5 (Experiments), the text states that “Video Quality is near saturation (flickering >92, smoothness >96).” While likely clear to experts, explicitly mentioning that these are scores on a 0–100 scale would prevent any potential ambiguity regarding the metric ranges. Additionally, the manuscript relies heavily on LaTeX macros (e.g., `\numvideo`, `\numturn`) for key statistics. While this is standard practice, the final review should verify that these expand correctly in the compiled PDF; if not, replacing them with explicit numerals in the text would improve immediate readability.

Finally, several figure captions in the Appendix (specifically `fig:qual_ee`, `fig:qual_sa`, and `fig:qual_ps`) appear to be empty or contain only the label text in the source. These should be populated with descriptive text explaining the qualitative examples shown (e.g., “Qualitative comparison of high-scoring vs. low-scoring event editing cases”) to ensure the appendix is self-contained and accessible without cross-referencing the main text. Addressing these minor points will further enhance the paper’s professional presentation.

Action items.

- **[writing]** In Section 5 (Experiments), the phrase ‘flickering >92’ and ‘smoothness >96’ lacks units or context. Explicitly state these are scores on the 0-100 scale to avoid ambiguity for readers unfamiliar with the specific metric ranges.
- **[writing]** The abstract and introduction use the macro `\numvideo` and `\numturn`. While standard in LaTeX, ensure the final compiled PDF renders these as the specific numbers (289 and 1,058) clearly. If the macro expansion is missing in the final output, replace with explicit numerals for immediate readability.
- **[writing]** In the Appendix, several figure captions (e.g., Fig. `qual_ee`, `qual_sa`, `qual_ps`) are empty or contain only the label text. Ensure these captions describe the visual content (e.g., ‘High vs. low scoring cases for event editing’) to make the appendix self-contained.